10

Modeling the Research Design

In Chapter 6 we discussed one way that we can use theory to offer a new understanding of an empirical result—reinterpreting by offering an alternative mechanism. In this chapter, we explore another approach. *Modeling the research design* involves representing a research design itself within a model that embodies the mechanisms in question. The goal is a better understanding of which theoretical implications are commensurable with the estimand of the research design.

In some sense, our approach to linking theory and empirics through all-else-equal claims suggests that much theorizing already implicitly involves a model of a research design. In particular, any comparative static can be thought of as a model of a just-controlling design with no omitted confounders. Modeling the research design becomes more interesting, and worth consideration as a distinct approach, when multiple mechanisms interact in potentially complicated ways or when a more complicated research design makes thinking about commensurability conceptually difficult.

In such circumstances, studying an analog of the research design within a theoretical model can be of help. This involves an additional step of theorizing beyond simply making a model of the phenomenon under study, but it is a step that can be particularly illuminating if the goal is to construct theoretical models with implications that speak very directly to empirical research. The key is to build a model that is both plausibly similar to the world and in which we can faithfully represent the research design.

10.1 Close Elections and the Incumbency Advantage

In Chapter 9, we discussed several mechanisms that help explain incumbent success in elections—party match, electoral selection, and insulation. We also

discussed Fowler's (2016) empirical strategies for disentangling these mechanisms. A key step in the disentangling involved using a regression discontinuity design to separate the insulation mechanism from the electoral selection mechanism.

This election regression discontinuity approach was pioneered by Lee (2008), who studied the effect of incumbency status on Democratic vote share in US House elections. The running variable in such an analysis is the Democratic share of the two-party vote in the previous election. If it is just below 50%, the winner is a Republican. If it is just above 50%, the winner is a Democrat. So in the subsequent election, whether the incumbent is a Democrat or Republican depends on where the running variable lies relative to this 50% threshold. Such a regression discontinuity design returns an estimate of the difference in Democratic vote share in districts where the Democrat barely won versus barely lost the previous election. Under the assumption that potential Democratic vote shares are continuous in lagged Democratic vote share at the 50% threshold, such a difference cannot be the result of other systematic differences between the winners and losers of the preceding election.

A common claim is that the estimates from such a regression discontinuity design isolate the insulation mechanism. For example, Erikson and Titiunik (2015, p. 102), in their discussion of the personal incumbency advantage, interpret the estimand from an election regression discontinuity as

> the votes a candidate gains upon becoming an incumbent due to the direct and indirect benefits of office-holding.

Similarly, Fowler and Hall (2014, p. 507) characterize the personal incumbency advantage as the answer to the question

> Holding all factors fixed, including the underlying quality and talent of the candidate and the incumbency status of the party, how much better will she perform in an election as an incumbent compared to the counterfactual scenario where she is a non-incumbent?

In our terminology, these quotes suggest that finding a nonzero incumbency effect in a regression discontinuity is a distinguishing implication of the insulation mechanism relative to other mechanisms (say, party match or electoral selection). Or, more specifically, in a model embodying both the insulation

mechanism and some other mechanism, the claim is that the election RD yields an estimate that is commensurable with the effect of insulation on reelection purged of all other effects.

While the claim that the regression discontinuity isolates (or, is commensurable with) the pure insulation mechanism is a common one, Eggers (2017) shows that assumptions above and beyond the usual RD assumption of continuity of potential outcomes are required to justify this interpretation. As we alluded to in Chapter 9, continuity of potential outcomes at the election threshold implies that, in the limit, winners and losers of close elections do not differ systematically from one another in terms of individual characteristics. For some quantities—for example, the effect of incumbency on the decision to run again—this is enough. But for commensurability with a pure insulation effect, we need more. The all-else-equal condition for commensurability with the pure insulation effect is that winners of close elections must not differ systematically (other than through their incumbency status) from the challengers they might face in subsequent elections.[1] If they do differ systematically, then the RD is not commensurable with pure insulation—a non-zero RD estimate reflects a combination of insulation and those other systematic differences. By modeling the research design, Eggers shows that, even in the RD sample, incumbents and challengers do indeed systematically differ, except under a knife-edge condition, so that the RD estimand does not isolate the insulation mechanism.

We illustrate Eggers's argument with a model inspired by his. There are many districts. Each district has two elections—first an open-seat election and second an election in which the incumbent from the first election runs against a new challenger. There is always a candidate from each party (L and R) in each election, and the incumbent always seeks reelection.

Each politician has a characteristic that some voters care about. To keep things as simple as possible, each politician is either high quality ($\overline{\theta}$) or low quality ($\underline{\theta}$). A politician is high quality with probability $p \in (0, 1)$, and qualities are independent across candidates. Candidates are randomly paired against each other in elections.

Voters observe candidates' quality perfectly. Each district has a unit mass of voters, divided into two kinds. To capture insulation, we assume there is a group of voters who always vote for the incumbent when one is available and otherwise vote for whichever candidate is of the higher quality. This group of

1. As we mentioned in Chapter 9, scare-off effects are part of the insulation mechanism. Scare-off plays no role in the discussion that follows.

		R Candidate: High	*R* Candidate: Low
L Candidate	High	$\frac{1}{2}, \frac{1}{2}$	1, 0
	Low	0, 1	$\frac{1}{2}, \frac{1}{2}$

FIGURE 10.1. Vote shares in an open-seat election. In each cell, the first number is the vote share of the *L* candidate and the second number is the vote share of the *R* candidate.

		R Candidate: High	*R* Candidate: Low
L Candidate	High	p^2	$p(1-p)$
	Low	$(1-p)p$	$(1-p)^2$

FIGURE 10.2. Electoral match frequencies in open-seat elections.

voters is of size λ. The remaining $1 - \lambda > 0$ voters vote for the higher quality candidate in all elections. When a voter is indifferent, she flips a coin.

The vote shares in an open-seat election in this model are in Figure 10.1. When both candidates are of the same quality, they win an equal number of votes; when the candidates are of different quality, the high-quality one gets the support of every voter.

A regression discontinuity design focuses on districts that had close open-seat elections and compares the vote share for one party in districts that just barely do and do not have incumbents from that party. Close elections are represented in the model by those situations where the voters split their votes evenly between the two candidates, that is, open-seat elections where the left- and right-party candidates were the same quality. These elections are the diagonal elements in Figure 10.1.

To start to see the problem that Eggers identifies, think about the distribution of high- and low-quality types among incumbents in districts that had a close open-seat election. The frequency of different open-seat match-ups is represented in Figure 10.2. There is a high-quality incumbent following a close open-seat election if and only if the open seat election matched two high-quality candidates against each other. And there is a low-quality incumbent following a close open-seat election if and only if the open seat election matched two low-quality candidates against each other. (It wouldn't have been a close election unless the two candidates were of the same quality.) So, the distribution of quality among incumbents in the RD sample is

$$\Pr(\overline{\theta} \mid \text{inc}, \text{RD}) = \frac{p^2}{p^2 + (1-p)^2}$$
$$\Pr(\underline{\theta} \mid \text{inc}, \text{RD}) = \frac{(1-p)^2}{p^2 + (1-p)^2}. \tag{10.1}$$

There is an important point to notice here. Conditional on there being a close open-seat election, it is true that the candidates *in that election* are not different from one another. But this is not the condition we need to isolate insulation. Rather, we need that the incumbents in the RD sample do not systematically differ from their challengers *in the next election*. And this, quite generally, is not the case.

The probability a challenger is high quality is just p. But in the RD sample, the probability an incumbent is high quality is determined by the number of close elections that involved two high-quality candidates compared to the number that involved two low-quality ones: $\Pr(\overline{\theta} \mid \text{inc}, \text{RD}) = \frac{p^2}{p^2+(1-p)^2}$. Comparing these, we find that incumbents in the RD sample are more likely to be high quality than are their challengers if $p > \frac{1}{2}$ and are less likely to be high quality than are their challengers if $p < \frac{1}{2}$. The RD sample of incumbents has the same distribution of quality as the pool of challengers if and only if $p = \frac{1}{2}$—that is, only under a knife-edge symmetry condition. Hence, for any $p \neq \frac{1}{2}$ the distribution of incumbent quality in the RD sample differs systematically from the distribution of challenger quality. The fact that each open-seat election in the RD sample is determined by a coin flip doesn't mean that restricting attention to close elections purges systematic differences between incumbents and future challengers.

What is going on? Voters flip coins only when the candidates have the same quality. If there are more high-quality than low-quality types in the population of candidates ($p > \frac{1}{2}$), then most of those coin-flip elections are between two high-quality candidates. As a result, the RD sample overrepresents high-quality candidates. If there are fewer high-quality than low-quality types in the population of candidates ($p < \frac{1}{2}$), then most of those coin-flip elections are between two low-quality ones. As a result, the RD sample overrepresents low-quality types. In either case, the regression discontinuity does not achieve what is needed to isolate insulation.[2]

2. Eggers shows that this result extends beyond the case of two quality types.

To see how this matters, it will be helpful to ask exactly what it would mean to isolate insulation. The ideal way to isolate insulation would be to experimentally randomize incumbency status. Under randomization, the distribution of abilities among incumbents is the same as the distribution of abilities among candidates in general (and therefore, among challengers).

What would the incumbent's expected vote share in the second election be in this ideal experiment? The incumbent would win the support of the λ voters who always support incumbents. And, since the abilities of incumbents and challengers would have the same distribution, on average the incumbent would win half the votes of the remaining $1 - \lambda$ voters who vote based on quality. As such, the incumbent's expected vote share under random assignment of incumbency is

$$\lambda + (1 - \lambda) \cdot \frac{1}{2}.$$

The expected vote share of any candidate in the open seat election is $\frac{1}{2}$. Call the extra votes a candidate expects to win by virtue of being an incumbent the *incumbency differential*. Under random assignment of incumbency, the incumbency differential is

$$\lambda + (1 - \lambda) \cdot \frac{1}{2} - \frac{1}{2} = \frac{\lambda}{2}. \tag{10.2}$$

So, if the regression discontinuity isolates the insulation effect, we should find that the incumbency differential is equal to $\frac{\lambda}{2}$ in our model with a first election rather than random assignment of incumbency. Let's see if it does.

Write $\mathbb{E}[\text{VS} \mid \theta, \text{inc}]$ for the expected vote share of an incumbent of type θ. The expected vote share of an incumbent in the RD sample is

$$\Pr(\overline{\theta} \mid \text{inc}, \text{RD}) \cdot \mathbb{E}[\text{VS} \mid \overline{\theta}, \text{inc}] + \Pr(\underline{\theta} \mid \text{inc}, \text{RD}) \cdot \mathbb{E}[\text{VS} \mid \underline{\theta}, \text{inc}]. \tag{10.3}$$

We calculated $\Pr(\overline{\theta} \mid \text{inc}, \text{RD})$ and $\Pr(\underline{\theta} \mid \text{inc}, \text{RD})$ in Equation (10.1). The other two terms are straightforward. Regardless of the incumbent's quality, the λ voters who value incumbency vote for her in the second election. If the incumbent is high quality, of the $1 - \lambda$ remaining voters, all vote for her in the event that the challenger is low quality and half vote for her in the event that the challenger is high quality. So the expected vote share of a high-quality incumbent is

$$\begin{aligned}\mathbb{E}[\text{VS} \mid \overline{\theta}, \text{inc}] &= \lambda + (1-\lambda)\left((1-p)\cdot 1 + p\cdot\frac{1}{2}\right)\\ &= \lambda + (1-\lambda)\left(1-\frac{p}{2}\right).\end{aligned}$$

If the incumbent is low quality, of the $1-\lambda$ remaining voters, half vote for her in the event that the challenger is low quality and none vote for her in the event that the challenger is high quality. So the expected vote share of a low-quality incumbent is

$$\begin{aligned}\mathbb{E}[\text{VS} \mid \underline{\theta}, \text{inc}] &= \lambda + (1-\lambda)\left((1-p)\cdot\frac{1}{2} + p\cdot 0\right)\\ &= \lambda + (1-\lambda)\frac{1-p}{2}.\end{aligned}$$

Substituting these into Equation (10.3), we find that the expected vote share of an incumbent in the RD sample is

$$\begin{aligned}&\lambda + (1-\lambda)\left(\frac{p^2}{p^2+(1-p)^2}\left(1-\frac{p}{2}\right) + \frac{(1-p)^2}{p^2+(1-p)^2}\left(\frac{1-p}{2}\right)\right)\\ &\qquad = \lambda + (1-\lambda)\left(\frac{1-3p+5p^2-2p^3}{2-4p+4p^2}\right).\end{aligned}$$

And since the expected vote share of any candidate in the open seat election is $\frac{1}{2}$, the incumbency differential in the RD sample is

$$\lambda + (1-\lambda)\left(\frac{1-3p+5p^2-2p^3}{2-4p+4p^2}\right) - \frac{1}{2}. \tag{10.4}$$

Comparing Equations (10.2) and (10.4), it is clear that the regression discontinuity isolates the insulation mechanism only if the term $\frac{1-3p+5p^2-2p^3}{2-4p+4p^2}$ is equal to $\frac{1}{2}$. We plot the incumbency differential from the RD as a function of p in Figure 10.3. Not surprisingly, as we've already discussed, the RD incumbency differential equals $\frac{\lambda}{2}$ only if $p = \frac{1}{2}$. (Recall we assume there are at least some high- and low-quality types, so p is not equal to 0 or 1.) For any other distribution of high- and low-quality types, the RD returns an estimate made up of a combination of the insulation mechanism and systematic differences in quality between incumbents and future challengers.

How should we think about this kind of result? It tells us that the regression discontinuity isolates the insulation effect only under a strong symmetry

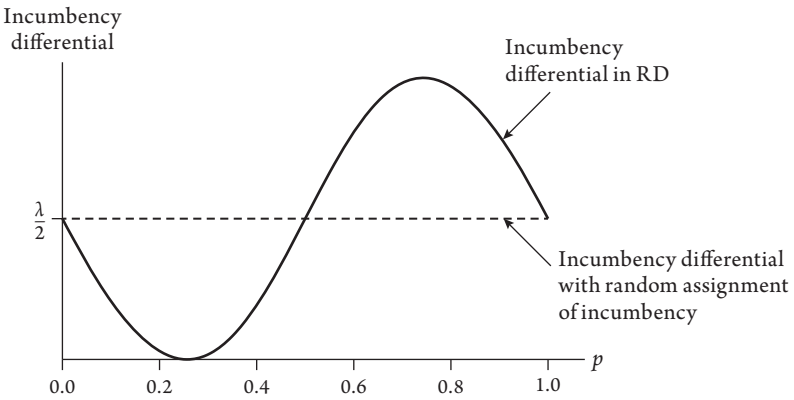

FIGURE 10.3. The RD isolates the insulation mechanism only if $p = \frac{1}{2}$.

assumption. So, if we make that symmetry assumption and it is representational, then a pertinent implication of our model is that the RD is commensurable with the pure insulation effect. But if we either don't make that symmetry assumption or don't believe it is representational, then we are not entitled to interpret the RD as isolating the insulation effect. One could also imagine separately assessing whether symmetry is representational. In this sense, Eggers's model of the research design offers some of the same benefits as a structural approach that provides "rigorous consideration of the theoretical assumptions which need actual empirical validation to deliver sensible parameter estimates" (Bombardini and Trebbi, 2020, p. 397).

10.2 The Efficacy of Protests

Scholars of political behavior and social movements want to understand how mass actions, such as protests or riots, affect policy decisions taken by the government. The literature posits a number of mechanisms by which such mobilization might yield a government response. For example, a protest may affect policy decisions by communicating information about the level of antiregime sentiment in the population (Lohmann, 1993, 1994), contributing to an extortionary strategy to extract concessions (Tilly, 2003), working as a tool of accountability to unseat an unpopular government (Fearon, 2011), attracting media attention (Wasow, 2020), or disrupting policy implementation. Mobilization might also spark a backlash, leading the government to increase repression or decrease its willingness to make concessions.

Empirical work on the relationship between mass mobilization and government actions thus addresses at least two possible questions. How does an increase in protest behavior affect government policy? And which mechanisms drive any such relationship? Answering either question credibly is difficult because mass mobilization results from choices that are likely to be endogenous to the government's behavior. For example, mobilization might not be worthwhile if the government already shares the policy goals of the mobilizable population, while mobilization might be a good tactic if the government is not already supportive of those goals.

A popular strategy to deal with these endogeneity problems uses weather shocks as a source of variation in mobilization in an IV analysis (e.g., Madestam et al., 2013; Ritter and Conrad, 2016; Wasow, 2020). There are concerns you might have about the validity of these instruments—for example, in some settings weather may directly affect the government's ability to respond. But let's put these concerns aside and grant that these studies have done a pretty good job of holding all-else-equal. (They certainly do a better job on this dimension than the prior literature.)

A commensurability question still looms: what exactly do we mean theoretically when we ask about the effect of the protesters' mobilization on the government's behavior? And how does this relate to the estimands of the empirical research designs?

This question is complicated because the treatment is itself a behavior, and people change their behavior for reasons. This is a particularly important point for informational mechanisms, in which the government's belief about the level of antiregime sentiment is a mediator for the effect of protest on policy choices. The informational content of a change in the citizens' behavior may well depend on the reasons for these changes.

Bueno de Mesquita and Tyson (2020) model the research design to clarify what we learn about the mechanisms underlying the efficacy of protest from such studies. To see the idea, consider a model of protest with two players: a government and a group of citizens. The group first decides whether to protest and then the government responds.

The group has one of two types, $\theta \in \{\underline{\theta}, \overline{\theta}\}$. Think of the group's type as representing its level of antigovernment sentiment. The prior probability that $\theta = \overline{\theta}$ is $p \in (0, 1)$. There is a cost to protesting, which can take one of two values: $\underline{c} < \overline{c}$, each of which is equally likely. High costs can be thought of as representing, for example, bad weather.

The group protests if and only if $\theta > c$. Assume $\overline{\theta} > \bar{c} > \underline{\theta} > \underline{c}$, so that a high type always protests and a low type protests if and only if the costs of protesting are low. This represents the idea that people who are more strongly antigovernment are willing to bear higher costs to protest that government.

The government observes protest behavior and forms a posterior belief, $\hat{p}$, about the probability that $\theta = \overline{\theta}$. The government's response is described by a function r that takes as an input whether or not there was a protest and the government's posterior belief. Let $\mathbb{I}$ be an indicator function that takes the value 1 if there was a protest and 0 if there was not. The government's response function is as follows:

$$r(\mathbb{I}, \hat{p}) = \begin{cases} \overline{\alpha} + \overline{\beta} \cdot \left(\hat{p}\right)^{\gamma} & \text{if } \mathbb{I} = 1 \\ \underline{\alpha} + \underline{\beta} \cdot \left(\hat{p}\right)^{\gamma} & \text{if } \mathbb{I} = 0, \end{cases}$$

with $\overline{\alpha} > \underline{\alpha} > 0$, $\overline{\beta} \geq \underline{\beta} > 0$, and $\gamma > 0$.

What does this response function represent? We can break it down into two components. Start by imagining moving from no protest $\mathbb{I} = 0$ to a protest $\mathbb{I} = 1$, but hold fixed the government's beliefs about antiregime sentiment at $\hat{p}$. The government's response changes by $\overline{\alpha} - \underline{\alpha} + (\overline{\beta} - \underline{\beta})\left(\hat{p}\right)^{\gamma}$. This is the government's *direct response* to protestors in the streets—for instance, holding fixed their beliefs about what citizens think, the government may be willing to make concessions or engage in repression to end a disruptive protest.

Second, imagine changing the government's beliefs about antiregime sentiment, say from $\hat{p}_0$ to $\hat{p}_1$, but without actually changing whether there was a protest. If we assume there is no protest, this will change the government's response by $\underline{\beta}(\hat{p}_1^{\gamma} - \hat{p}_0^{\gamma})$. And if we assume there is a protest it will change the government's response by $\overline{\beta}(\hat{p}_1^{\gamma} - \hat{p}_0^{\gamma})$. This is the government's *informational response*—for instance, the government's willingness to make concessions or engage in repression may depend on how disgruntled it believes its citizens are. The parameter γ allows flexibility in the shape of the informational response. (If $\gamma < 1$, the function is concave, if $\gamma = 1$ it is linear, and if $\gamma > 1$ it is convex.)

Finally, the comparison of $\overline{\beta}$ and $\underline{\beta}$ determines whether or not the direct and informational responses are additively separable. If $\overline{\beta} = \underline{\beta} \equiv \beta$, then the direct response is just $\overline{\alpha} - \underline{\alpha}$, which doesn't depend on the government's beliefs at all. And the informational response is $\beta(\hat{p}_1^{\gamma} - \hat{p}_0^{\gamma})$, regardless of

whether or not there was a protest. By contrast, if $\overline{\beta} > \underline{\beta}$, then the direct and informational responses are not additively separable. The size of the direct response depends on the government's beliefs and the size of the informational response depends on whether there was a protest. The responses might not be additively separable because, for example, the government might be more willing to engage in repression to end a disruptive protest when it is particularly convinced citizens are highly disgruntled.

10.2.1 Representing the Ideal Experiment: Unobservable Costs

What does the literature have in mind when it asks about the efficacy of protesting? Presumably, it wants to know something like whether protests are an effective strategy for a group of citizens attempting to change government policy. In other words, are a group of aggrieved citizens more likely to achieve their goals if they protest than if they don't?

Bueno de Mesquita and Tyson argue that, inside the model, the ideal experiment for an empiricist trying to answer this question involves a shock to some feature of the world that influences citizens' protest behavior, but that is not observable by the government. Such unobservable shocks induce a change in the citizens' behavior whose reason is unknown by the government. As such, from the government's perspective, such a change in behavior might well result from a genuine shift in antigovernment sentiment. Hence, such unobservable shocks activate both the direct and informational mechanisms associated with protest in a way that gets at the effects that are of interest for substantive questions about the efficacy of protest.

To get at the key commensurability question, they first represent this ideal experiment in the model and derive the implications. They then represent actual research designs used in the literature and compare their implications to the implications of the ideal experiment.

To represent the ideal experiment, consider a situation in which c cannot be observed by the government but can be observed by the citizen group.

Let's start by calculating the government's posterior beliefs. Using Bayes' rule, if a protest occurs, the government has beliefs:

$$\Pr(\overline{\theta} \mid \text{protest}) = \frac{2p}{1+p}.$$

If a protest does not occur, the government is certain the group is a low type:

$$\Pr(\overline{\theta} \mid \text{no protest}) = 0.$$

What is the average effect on the government's response of a decrease in the cost of protesting?

If costs are low, then a protest occurs for sure and the government's response is $r\left(1, \frac{2p}{1+p}\right)$. If costs are high, then a protest occurs with probability p. If a protest occurs, the government's response is again $r\left(1, \frac{2p}{1+p}\right)$. But if a protest does not occur, the government's response is $r\,(0,0)$. So the average effect of reducing costs is

$$\begin{aligned} & r\left(1, \frac{2p}{1+p}\right) - \left[pr\left(1, \frac{2p}{1+p}\right) + (1-p)r\,(0,0)\right] \\ & = (1-p)\left[\overline{\alpha} - \underline{\alpha} + \overline{\beta} \cdot \left(\frac{2p}{1+p}\right)^{\gamma}\right]. \end{aligned} \tag{10.5}$$

Equation (10.5) represents the reduced-form causal relationship between the costs of protesting and the government's behavior. (This is analogous to the reduced form in our discussion of instrumental variables in Chapter 5.) Of course, lowering the cost does not always result in a change in protest behavior. With low costs there is always protest, whereas with high costs, there is protest with probability p. Consequently, a shock that lowers the costs has effect $1-p$ on the probability of protest. Dividing the reduced-form relationship in Equation (10.5) by this first stage gives the local total average effect of increased protest on government response that results from lowering the cost of protest in a way that is unobservable to the government (this is the theoretical analog of the IV estimand). We label this τ_u:

$$\tau_u = \overline{\alpha} - \underline{\alpha} + \overline{\beta} \cdot \left(\frac{2p}{1+p}\right)^{\gamma}. \tag{10.6}$$

We can decompose this total effect into two substantive components: a direct effect and an informational effect. The direct effect of an increase in protest behavior is the effect of increasing protest behavior holding beliefs fixed (here we hold them fixed at $\frac{2p}{1+p}$):

$$\delta_u = \overline{\alpha} - \underline{\alpha} + \left(\overline{\beta} - \underline{\beta}\right) \cdot \left(\frac{2p}{1+p}\right)^{\gamma}. \tag{10.7}$$

The informational effect is the effect of changing the government's beliefs from what they would be if no protest occurs (0) to what they would be if

a protest occurs ($\frac{2p}{1+p}$), holding protest behavior fixed (here, fixed at a protest not occurring):

$$\iota_u = \underline{\beta}\left(\frac{2p}{1+p}\right)^{\gamma}. \tag{10.8}$$

The total effect (τ_u) is equal to the sum of the direct effect (δ_u) and informational effect (ι_u). Notice, we could have defined the direct effect holding beliefs fixed instead at 0 and then redefined the information effect of a protest occurring. Nothing hinges on this choice.

Importantly, τ_u, δ_u, and ι_u do not necessarily represent quantities whose analogs are observable by an empirical researcher. Rather, these are representations of effects of theoretical interest in the ideal experiment, which an empirical researcher might never encounter. The key question is whether some actual research design generates estimates that are commensurable with any of these.

10.2.2 Representing an Actual Research Design: Observable Costs

A research design using weather as an instrument differs from the ideal experiment because weather is observable to the government. So the next step in modeling the research design is to study the effect of an observable shock to the cost of protesting. Then we assess commensurability by asking whether the theoretical quantity that represents the estimand of the actual research design corresponds to a theoretical quantity of interest from the ideal experiment.

So let's think about the same model, but now assume that the costs are observable to the government. Using Bayes' rule again, we now have to calculate the government's beliefs conditional on both whether or not a protest occurs and whether costs are high or low. If costs are low, both types protest, so the government learns nothing and its posterior equals its prior:

$$\Pr(\overline{\theta} \mid \text{protest}, \underline{c}) = p.$$

By contrast, if costs are high, one type protests and the other doesn't, so the government learns everything:

$$\Pr(\overline{\theta} \mid \text{protest}, \overline{c}) = 1.$$

$$\Pr(\overline{\theta} \mid \text{no protest}, \overline{c}) = 0.$$

What is the average effect on the government's response of the increased probability of protest associated with a decrease in the cost of protesting when costs are observable?

If costs are low, then a protest occurs for sure and the government's response is $r\left(1,p\right)$. If costs are high, then a protest occurs with probability p. If a protest occurs, the government's response is $r\left(1,1\right)$, but if a protest does not occur, the government's response is $r\left(0,0\right)$. So the average effect of reducing observable costs is

$$r\left(1,p\right)-\left[pr\left(1,1\right)+\left(1-p\right)r\left(0,0\right)\right]=\left(1-p\right)\left(\overline{\alpha}-\underline{\alpha}\right)+\overline{\beta}\left(p^{\gamma}-p\right).$$

Just as before, we divide this reduced form by the first stage $\left(1-p\right)$ to get the effect of the change in protest behavior due to a change in costs $\left(\tau_o\right)$:

$$\tau_o=\overline{\alpha}-\underline{\alpha}+\overline{\beta}\cdot\frac{p^{\gamma}-p}{1-p}.$$

If an empirical researcher uses a research design that corresponds to this nonideal experiment (i.e., one with observable cost shocks), then τ_o is commensurable with the resulting IV estimate.

10.2.3 Comparing the Actual Research Design and the Ideal Experiment

It is now straightforward to think about what we learn from weather shocks about the mechanisms underlying the efficacy of protest. An empirical researcher using a research design with observable cost shocks observes an estimate that is commensurable with τ_o. But the theoretical quantity most closely related to substantive questions about the overall efficacy of protest is τ_u. Also of interest are the theoretical quantities decomposing the two mechanisms, δ_u or ι_u. The question is whether the research design's estimand is commensurable with any of these. Since we know the research design's estimand is commensurable with τ_o, we can figure this out by asking whether τ_o is equal to any of the quantities of theoretical interest from the ideal experiment.

A first observation is that the total effect of the change in protest induced by a shock to costs is not the same when that shock is observable as when it is not. One can see this by checking whether $\tau_o=\tau_u$. They are equal if and only if

$$\left(\frac{2p}{1+p}\right)^{\gamma}=\frac{p^{\gamma}-p}{1-p},$$

which never holds. The left-hand side is greater than the right-hand side for any $p \in (0, 1)$ and $\gamma > 0$. This means that the research design using observable weather shocks is not commensurable with the total effect in the ideal experiment (i.e., an experiment with unobservable shocks).

Why do the actual research design and the ideal experiment yield different total effects for the same shock to costs? In both cases, the shock induces the same change in protest behavior—the probability of protest increases by $1 - p$. But there are different effects of that change on the government's beliefs in the actual research design and the ideal experiment. When costs are observable, the informational content of a protest depends on what the costs are—the government understands that a large protest means something different on a sunny day and on a rainy day. When costs are low, there is no informational content—holding a protest on a nice sunny day is too easy for the government to conclude anything about citizen attitudes. But when costs are high, there is considerable informational content because only deeply aggrieved citizens protest in the rain. By contrast, with unobservable shocks the government has to average across these possibilities, leading to different beliefs in the two scenarios.

We can see that the same change in average protest behavior has different effects on government behavior in the ideal experiment (represented in the model with unobservable shocks) and the actual research design (represented in the model with observable shocks). As a result, the model indicates that standard research designs are not commensurable with the total effect of protest in the ideal experiment.

A thought you might have is that, while the estimate generated by a research design with observable shocks does not correspond to the total effect of increased protest in the ideal experiment, maybe it corresponds to just the direct effect in the ideal experiment. The intuition is that when the government observes the cost shock, it knows those shocks are causing a change in protest behavior. Hence, it won't mistakenly think that increased protest size due to lower costs is the result of a genuine change in antigovernment sentiment. So, maybe, once it takes the shock into account, there is no informational content left in the change in protest behavior. If this is the case, perhaps the effect being estimated by a research design using observable shocks to protest costs captures the direct effect in the ideal experiment, purged of any informational effect. That is, maybe this research design distinguishes the direct mechanism from the informational mechanism.

We can explore this possibility in our model of the research design as well. Comparing the total effect in the model with observable shocks (representing the quantity estimated by the actual research design) to the direct effect in the model with unobservable shocks (i.e., the direct effect in the ideal experiment), and rearranging, we see that $\tau_o = \delta_u$ if and only if

$$\frac{(1+p)^{\gamma}(p^{\gamma}-p)}{(2p)^{\gamma}(1-p)} = \frac{\overline{\beta}-\underline{\beta}}{\overline{\beta}}.$$

The left-hand side of this condition is strictly decreasing in γ. The right-hand side is constant in γ. As such, for any $p \in (0, 1)$, this condition holds for at most one value of γ, which is to say that the estimand of the actual research design is commensurable with the direct effect in the ideal experiment only in a knife-edge condition.

To start to get some intuition for what is going on, think about the case in which the direct and informational effects of protest on government behavior are additively separable (i.e., $\overline{\beta} = \underline{\beta}$). In this case, the right-hand side is zero and the unique γ for which the condition holds is $\gamma = 1$. That is, if the direct and informational effects of protest on government behavior are additively separable, then the total effect in the actual research design equals the direct effect in the ideal experiment if and only if the government's behavior is linear in its beliefs. Why is this true?

Additive separability implies that the direct effect is the same regardless of beliefs. This means that the direct effect in the actual research design and the ideal experiment are the same, even though the government always has different beliefs in those two settings. As such, for the total effect in the actual research design to equal the direct effect in the ideal experiment, there must be no information effect in the actual research design. It might seem that this is impossible, since the government does learn from observing protest size. But, as must be the case for a Bayesian actor, on average the government's posterior beliefs equal its prior beliefs.[3] And so the information effect, which averages across the possible shocks, is zero if and only if beliefs enter linearly, so that the average is all that matters.

If there is not additive separability (i.e., $\overline{\beta} \neq \underline{\beta}$), then there is in fact heterogeneity in the direct effect itself, since it depends on the government's beliefs.

3. This is the Law of Iterated Expectations.

So now the direct effects in the actual research design and the ideal experiment are not equal. As such, for the total effect in the actual research design to equal the direct effect in the ideal experiment, it must be that the information effect in the actual research design is of precisely the right size and sign to exactly offset the difference between the direct effects in the actual research design and in the ideal experiment. This is obviously a knife-edge condition, reflected in the fact that it holds for one and only one value of γ.

The key idea in this example is that rainfall is observable to the government just as it is to the analyst (and the protestors). Even though weather may be random, the government will potentially respond differently to protests that occur in good and bad weather, since a protest on a sunny day conveys different information about antigovernment sentiment than a protest on a rainy day. The estimated effect of a protest on government policy always reflects both the direct effect of the protest and the indirect effect of the protest that operates through the government's beliefs. Estimates from research designs that exploit weather shocks capture the effect on the government's beliefs inclusive of the government's ability to adjust its interpretation of the protest in light of the weather. Such estimates differ from the theoretical ideal experiment, which exploits an unobservable shock, where the government can't reinterpret the meaning of the protest in light of the shock (since the government doesn't know the shock occurred).

10.3 Gleanings

As we said at the outset of this chapter, any comparative static can be thought of as a model of a just-controlling research design. The papers discussed in this chapter show how modeling the research design can become more interesting and productive when the research design in question is itself more complex. As Eggers shows, to understand which components of the incumbency advantage are and are not disentangled by a regression discontinuity design, it is not sufficient to simply study the incumbency advantage in a model of elections. One needs to directly analyze the incumbency advantage among those politicians in the model who become incumbents in a prior close election. Similarly, Bueno de Mesquita and Tyson show that there is not a univocal concept of the effect of increased protest activity. That effect depends on the source of variation in protest behavior and, as such, it is productive to model and compare the effects of different sources of variation.

The model of the research design in each of these papers sheds light on whether it is possible to disentangle or distinguish some particular mechanisms. And, in both cases, it turns out that the required commensurability is present only under knife-edge conditions. Eggers shows that a regression discontinuity analysis disentangles insulation and electoral selection only if the distribution of politician quality types is perfectly symmetric. Bueno de Mesquita and Tyson show that an observable shock isolates the direct effect of protests only if beliefs enter the government's response in just the right way.

Finally, Bueno de Mesquita and Tyson point out that when the treatment of interest is itself endogenous behavior, responses to that treatment may depend on the source of the variation in behavior. So different research designs, which exploit different sources of variation, estimate different treatment effects. This is particularly relevant for thinking about the interpretation of experimental findings where the source of variation is manipulation of behavior that subjects may perceive as endogenously determined.